Neural SQL Analytics Architecture for Enterprise Data

Abstract & Introduction (Page 1)

This research introduces the Hybrid Dual-Encoder Architecture for natural language database interfaces. Our architecture integrates relational PostgreSQL schemas with high-dimensional dense vector embeddings. The primary goal is bridging semantic natural language requests with strict relational consistency.

Research Methodology (Page 2)

Our methodology consists of four core phases:

1. Schema Graph Extraction: Mapping relational primary and foreign key constraints into structured AST graphs.

2. Dense Retrieval Pipeline: Generating 768-dimensional normalized vectors via dual-encoder embeddings.

3. Hybrid Reranking: Combining cosine similarity distance with BM25 lexical token frequencies.

4. Safe Execution Sandbox: Enforcing read-only transactional boundaries and strict mutation confirmation gates.

Experimental Limitations (Page 3)

The primary limitation of this research is that latency scales logarithmically when the vector store exceeds 100,000 document chunks without GPU-accelerated IVFFlat indexes.
A secondary limitation is that non-English queries require cross-lingual alignment layers.